



Fig (7.1) (a)



Fig (7.1) (b)



Fig (7.1) (c)

A balloon kept under sunlight shattered, why? Why a hot coffee or tea in a cup became cold as the time passes? Clothes dry up quickly under sunlight? Honey is thicker than water, why?

Why do water and milk or other liquids boil at different temperatures? Why do water and milk take the shapes of the container in which they are poured? Have you ever think that when you sit at your chair or bed, their foams compresses but their wooden frame do not?

After studying this unit you will be able to find the answers of such questions and other similar questions and develop the clear concepts.

7.1 STATES OF MATTER

There are three states of matter. These states are solid, liquid and gas. All the material objects around us belong to any one of these states. Water is the best example of three states of matter.

- ◆ The solid state of water is ice figure. (7.1)(a). Ice exists in many forms like, ice cubes, snow, glaciers and icebergs.
- ◆ The liquid state is water itself figure. (7.1)(b). Water is found in oceans, rivers and underground deposits.
- ◆ The gaseous state of water is steam. The “white smoke” that you see in figure. (7.1)(c) is, in fact, a small cloud formed by water vapours in air above the cup.

These states have different properties which are listed in the following table.

Table (7.1) Properties of Matter

States of Matter	Shape	Volume	Density	Compressibility
Solid	Fixed	Fixed	High	Incompressible
Liquid	Not fixed	Fixed	High	Incompressible
Gas	Not fixed	Not fixed	Low	Compressible



Do You Know

Water is different from other substances because it is less dense in its solid state (ice), than its liquid state (water).

Addition or removal of a certain amount of energy can change the state of a matter. The terms for these changes in the state are:

Melting: conversion from solid to liquid.

Boiling: conversion from liquid to gas.

Condensing: conversion from gas to liquid.

Freezing: conversion from liquid to solid.

Evaporation: conversion from liquid to gas.

Evaporation is different from boiling.

Evaporation is a process by which a liquid becomes a gas at temperatures below its boiling point.

For example, drying of wet clothes, drying of wet floor etc. Conversion of matter between three states involves physical changes and not chemical changes.

Why liquids and gases take the shapes of their containers while solids have definite shapes? Why do different substances boil and melt at different temperatures? Why can gases be compressed easily while solids and liquids cannot?

The answers of the above and such other questions can be obtained by considering the arrangements of the particles in these states and how these particles are able to move about. This is explained by the kinetic molecular theory of matter.

Kinetic Molecular Model of Matter

The kinetic molecular model of matter is,

Matter is made up of tiny particles called atoms, or group of atoms called molecules. These molecules are always in continuous random motion.

According to this model particles are in continuous motion. Thus an alternative name for model is 'The particle model of matter'.

The evidence of molecular motion is Brownian motion.



Weblinks

Web link of Brownian motion.

<http://www.phyntau.edu>



Solid

Fig (7.2) (a)



Liquid

Fig (7.2) (b)

Brownian Motion

The evidence of molecular motion first discovered by the botanist Robert Brown in 1827. He observed the irregular motion of pollen grains suspended in water and deduced that the water molecules were in constant, random motion. This irregular motion caused by water molecules is called "Brownian motion" named after the scientist.

The kinetic molecular theory explains the physical properties of solids, liquids and gases by considering the position and motion of molecules.

The particles in **solids** (Fig7.2a) have following features:

- ◆ The molecules are closely packed together and occupy minimum space.
- ◆ The molecules usually arranged in a regular pattern called lattice.
- ◆ There is a large number of particles per unit volume. That is why solids have the highest densities.

The movement of particles in solids have following features:

- ◆ The forces of attraction between particles are very strong.
- ◆ The particles are not able to change positions.
- ◆ The particles vibrate about fixed positions thus are not entirely stationary.
- ◆ This explains why solids have fixed shapes and volumes.

The movement of particles in liquids (Fig 7.2b), have following features:

- ◆ The molecules are slightly further apart compared to that of solids.
- ◆ The molecules occur in clusters.

- ◆ There is slightly less number of particles per unit volume compared to solids.

◆ This is why liquids have relatively high densities. The movement of particles in liquids have following features:

- ◆ The forces of attraction between particles are strong.
- ◆ The particles are free to move about within the liquid.

◆ These features explain why liquids have fixed volumes, but take the shape of the container.

The particles in gases (Fig 7.2c), have following features:

- ◆ The molecules are very far apart.
- ◆ The molecules are arranged randomly and are free to move with very high speeds.

◆ There is small number of particles per unit volume. The movement of particles in gases has following features:

- ◆ The forces of attraction between particles are negligible.
- ◆ The particles are able to move freely in random directions at very high speeds.
- ◆ The particles occupy any available space.

SELF ASSESSMENT QUESTIONS:

- Q1: Explain why the measurement of volume of a given liquid remains same although it is measured by measuring cylinders of different shapes and sizes.
- Q2: What is the difference between evaporation and boiling?
- Q3: What is the difference between three states of matter? in terms of the spacing between the molecules.



Human body consists of all three states of matter.

1. Solid in the form of organs.
2. Liquid in the form of blood.
3. Gas in the form of Oxygen and carbon dioxide for respiration.



Gas
Fig (7.2) (c)



7.2 FORCES AND KINETIC THEORY

Why some materials are solid and liquid while others are gases at room conditions?

Forces between the molecules are responsible for the different states of matter as well as for the physical properties. According to the kinetic molecular model molecules of gases have large kinetic energy as a result there are no forces of attraction between them as a result molecules of gases can move freely and go farther apart. This is why gases can occupy any available space and can be compressed easily. Boiling and melting points of gases are also very low because of this reason. The molecules of liquids as compared to that of the gases have less kinetic energy hence intermolecular forces come into play. That is why the molecules of liquids are very close to each other but still free to move, therefore liquids do not have fixed shape but fixed volume. The melting and boiling points of liquids are also high as compared to gases. The molecules of solids have extremely lowest energies therefore experience strong attractive forces and can not move freely but only have small vibrations about mean positions. This gives solid a fixed shape and volume. That is why densities, melting and boiling points of solids are very high.

As a result we are able to convert water into ice, cream into ice cream, natural gas into compressed natural gas 'CNG' etc.

The state of a substance can be changed either by heating or by cooling it. on the other hand when a solid substance (Fig7.3) is heated, the molecules start to vibrate more and more strongly. Eventually, the molecules vibrate more violently and inter molecular forces become weak. As a result material becomes a

liquid, if process of heating is continued further, then molecules have sufficient energy to overcome all of the attractive forces as a result 'substance becomes a gas'.

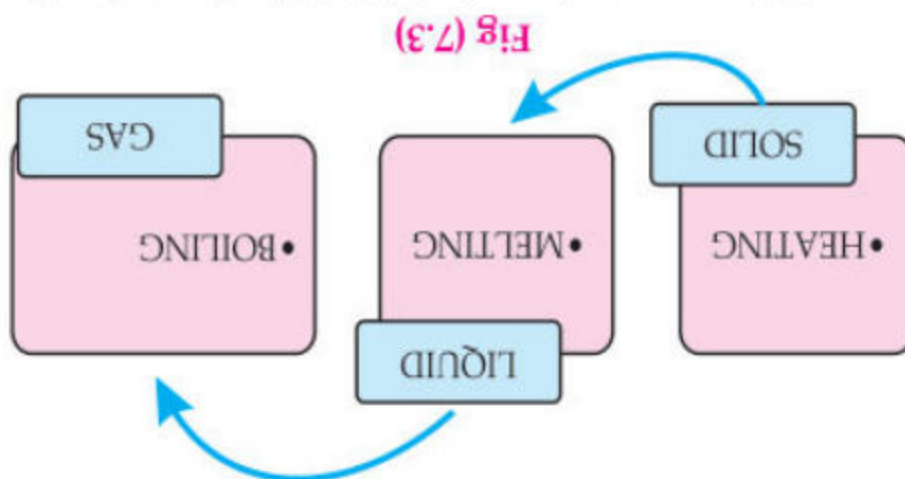


Fig. (7.3)

When a gas is cooled (Fig 7.4), the molecules move more slowly and collide with one another, may stick together and force of attraction between molecules increases. Keep cooling the gas and eventually all of the molecules stick together to form a liquid. Further cooling will cause all the molecules to stick together to form a solid.

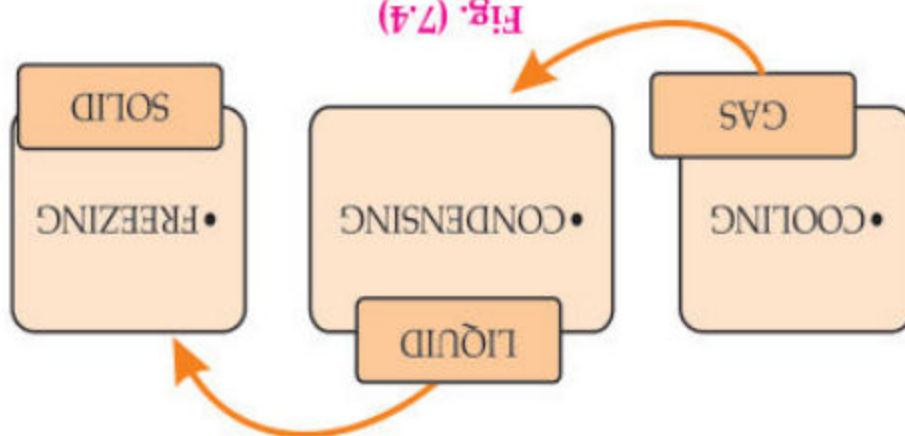


Fig. (7.4)

Table (7.2) on the right shows the boiling and melting points of some pure substances:

Helium has lowest boiling and melting points as compared to other substances. It solidifies only when it is cooled and compressed. Mercury is the only metal that is not solid at room temperature.

Substance	Melting point (°C)	Boiling point (°C)
Helium	-272	-269
Oxygen	-218	-183
Nitrogen	-191	-177
Mercury	-39	257
Water	0	100
Iron	2080	3570
Diamond (Carbon)	4100	5400
Tungsten	3920	6500

Table (7.2)
Boiling and melting points
of some pure substances





Self Assessment Questions:

- Q4: Why Tungsten melts at a much higher temperature than iron?
- Q5: What is the name of process in which a liquid changes into a solid?
- Q6: What is the name of temperature at which a liquid changes into a solid?

7.3 GASES AND THE KINETIC THEORY

Kinetic molecular theory clearly describes the properties and behavior of gases. Hot air balloons (Fig 7.5) are the practical applications of the discussion given below:

The behavior of gases

The molecules in the gases have relatively large distance between them. The molecules in the gases move about very quickly. A gas molecule moves in a straight line. It changes its direction only when (i) it collides with another gas molecule or (ii) with the walls of its container. After collision it moves away in a new direction. Since gas molecules collide many times each second. Therefore the motion of molecules is constant and random.

The behavior of a gas can be described completely by its pressure, volume and temperature.

Pressure

We already know that pressure is defined as the force per unit area. All the gases exert pressure on the walls of their container. This pressure is the total force exerted per unit area by the gas molecules during collision. The gas molecules exert pressure only when they collide with the walls. The number of collisions is proportional to the number of molecules. If the number

Fig (7.5)



of molecules is doubled then number of collisions will also be doubled (Fig 7.6 a, b, c and d). Hence the pressure is also doubled.

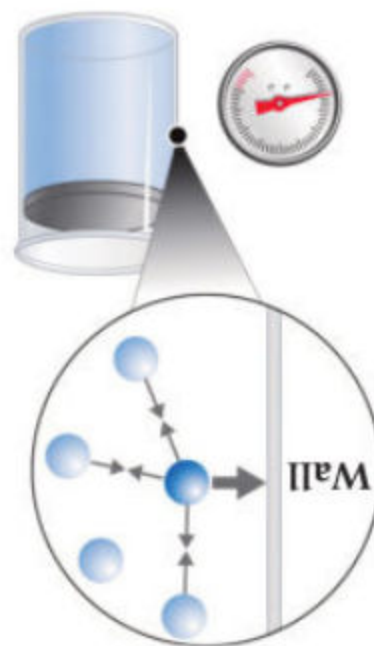


Fig. (7.6) (a)
Low pressure

of molecules is doubled then number of collisions will also be doubled (Fig 7.6 a, b, c and d). Hence the pressure is also doubled.

Blowing up a balloon is an example of pressure.

If more air is pushed into the balloon it will be inflated more. Because air molecules apply pressure on the rubber walls of balloon hence it gets inflated.

Pressure of a gas can also be increased by compressing it. This is done by reducing the size of the gas container (Fig 7.7). The gas molecules have been compressed into a smaller volume so they will collide more frequently with the walls of container and creates more pressure. If the gas is compressed to half its original volume its pressure will be doubled.

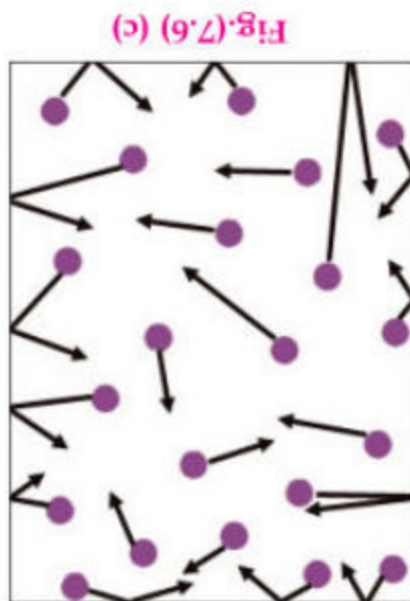


Fig. (7.6) (c)

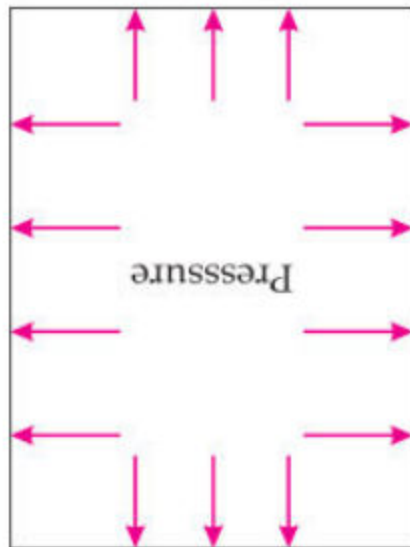


Fig. (7.6) (d)

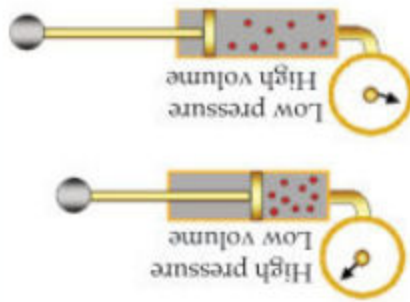
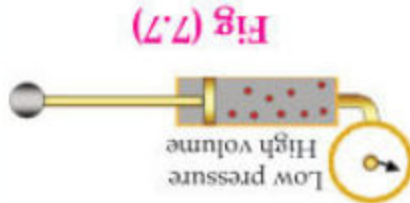


Fig (7.7)





Volume

We know that the space occupied by substance is known as volume. The gas has no definite volume because the molecules of the gas are far away from each other and can move freely at high speeds. Therefore gas always takes up the shape and volume of its container. For example, the smell of a perfume quickly spreads through the room as soon you spray it at your body or clothes. Because, the molecules move freely and randomly at high speeds through out the room.

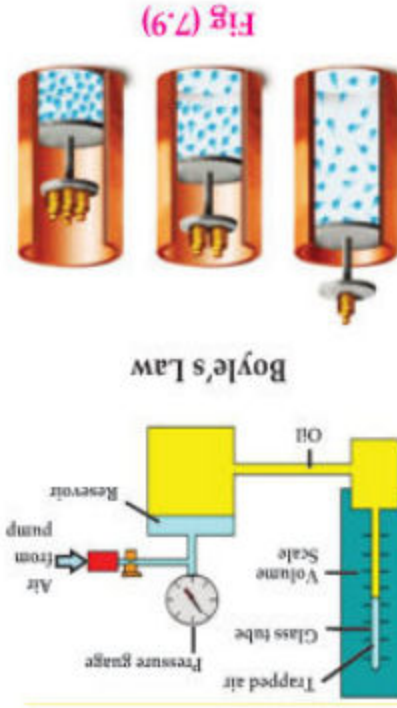
Volume of a gas can also be increased by decreasing its pressure. This could be done by reducing the load on the piston of the gas container. As the gas molecules are in random motion (Fig 7.8) they quickly covers the whole space and the volume increases. If the gas is compressed to half its original volume its pressure will be doubled.

Temperature

The temperature of a gas is determined by the average translational kinetic energy of its molecules. If a gas is heated the average translational kinetic energy of its molecules increases and temperature of the gas rises. If a gas is cooled down the average translational kinetic energy of its molecules decreases and temperature of the gas falls.

Pressure - volume relationship in gases

Robert Boyle, an English physicist and chemist in 1662, studied the relationship between pressure and volume of a gas (Fig 7.9). The results of a Boyle's experiment are shown below:



$$p \propto \frac{1}{V} \text{ or } V \propto \frac{1}{p}$$

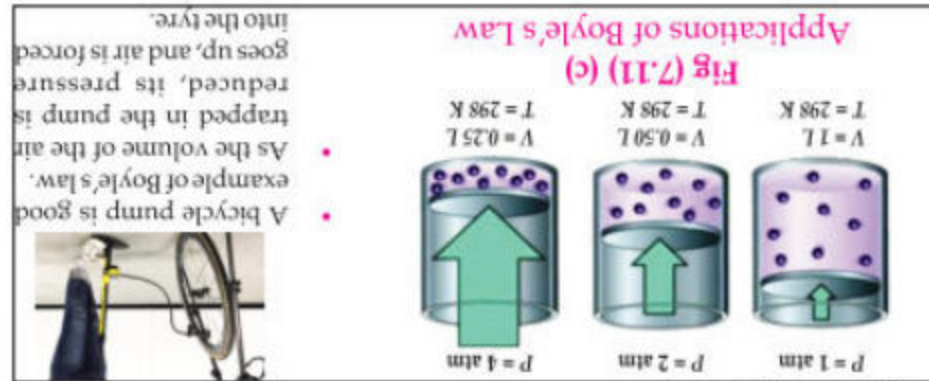


Fig. (7.11)(a, b, c):

Some applications of pressure-volume (p-V) relationship of a gas i.e. Boyle's law are given below in

Applications of (p-V) relationship of a gas "Boyle's law"

The volume of a fixed mass of a gas is inversely proportional to its pressure, provided its temperature remains constant.

law' which states that;

Thus Robert Boyle conclude his law known as 'Boyle's

$$p_1 V_1 = p_2 V_2 \dots \dots \dots (7.2)$$

pressure x final volume

can write Initial pressure x initial volume = Final

$$pV = \text{constant} \dots \dots \dots (7.1)$$

and volume is constant i.e.

5. At constant temperature the product of pressure

straight line passing through the origin.

4. The graph between "p" and "1/V" shows a

vice-versa, i.e.

3. The graph between "p" and "V" shows that if

and between "p" and 1/V" are shown below:

2. The graph (Fig.7.10) between "p" and "V"

volume becomes one-third and so on.

half. If pressure increases by three times then

1. If pressure of the gas is double its volume becomes



Fig (7.10)

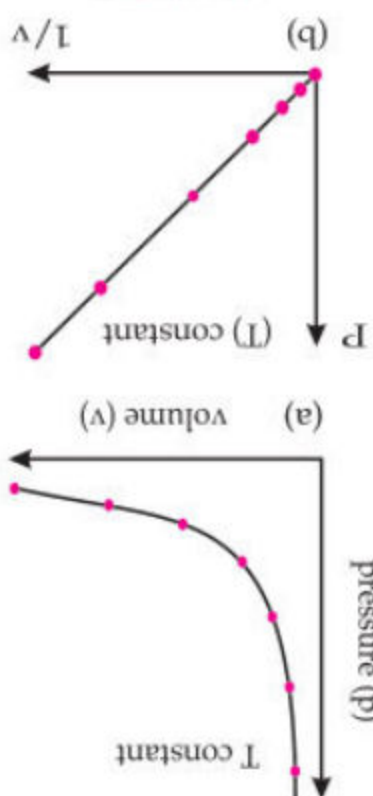


Fig (7.11) (a)



Fig (7.11) (b)

**Worked Example 1**

A cylinder contains 60cm^3 of air at a pressure of 140kPa . What will its volume be if the pressure on it is increased to 420kPa ?

Solution

Step 1: Write down the known quantities and quantities to be found.

$$P_1 = 140\text{kPa}$$

$$V_1 = 60\text{cm}^3$$

$$P_2 = 420\text{kPa}$$

$$V_2 = ?$$

Step 2: Write down the formula and rearrange if necessary.

$$P_1 V_1 = P_2 V_2 \text{ or}$$

$$V_2 = \frac{P_1 \times V_1}{P_2}$$

Step 3: Put the values and calculate.

$$V_2 = \frac{140\cancel{\text{kPa}} \times 60\text{cm}^3}{420\cancel{\text{kPa}}} = 20\text{cm}^3$$

The new volume is 20cm^3 .

Worked Example 2

Air at a pressure of $1.0 \times 10^5\text{Pa}$ is contained in a cylinder fitted with a piston. The air is now compressed by pushing the piston, so that the same mass of air now occupies one-fifth the original volume without any change in temperature. Calculate the pressure of the air.

Solution

Step 1: Write down known quantities and quantities to be found.

**Do You Know!**

An important feature of the equation $p_1 V_1 = p_2 V_2$ is that it does not matter what units we use for p and V , as long as we use the same units for both values of p (for example Pa, kPa or atmosphere etc), and the same units for both values of V (for example m^3 , dm^3 or cm^3 etc)

$$P_1 = 1.0 \times 10^5 \text{ Pa}$$

$$V_1 = V_1 \text{ cm}^3$$

$$V_2 = \frac{1}{5} V_1 \text{ cm}^3$$

Step 2: Write down the formula and re arrange if necessary.

$$P_1 V_1 = P_2 V_2 \text{ or}$$

$$P_2 = \frac{P_1 \times V_1}{(1/5)V_1}$$

Step 3: Put the values in formula and calculate.

$$P_2 = \frac{1.0 \times 10^5 \text{ Pa} \times V_1 \text{ cm}^3}{(1/5)V_1 \text{ cm}^3} = 5.0 \times 10^5 \text{ Pa}$$

So, the final pressure is now $5.0 \times 10^5 \text{ Pa}$.

Self Assessment Questions:

Q7: Draw diagrams of the molecules in a gas to explain the effect of pressure change on its volume.

Q8: What is the meant by the subscripts (1) and (2) in the equation, $P_1 V_1 = P_2 V_2$?

Q9: What is the effect of temperature on average translational kinetic energy of molecules?

**Do You Know!**

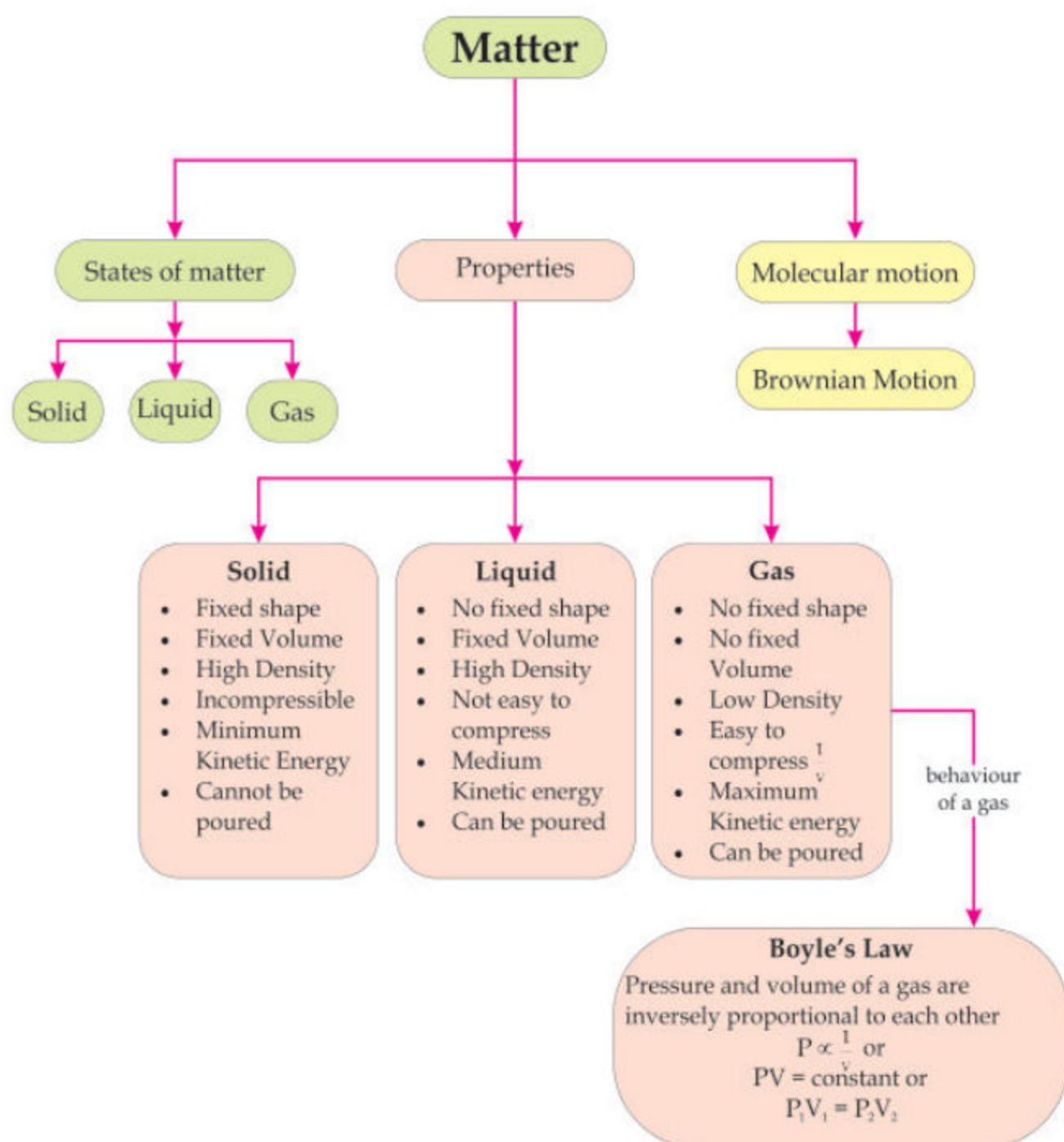
when gas is compressed, volume decreases and the pressure increases.

**SUMMARY**

- ◆ Matter exists in three states: solid, liquid and gas.
- ◆ The state of a matter can be changed by adding or removing a certain amount of energy.
- ◆ The kinetic molecular theory is based upon the arrangement and movement of molecules in a substance.
- ◆ The kinetic molecular theory suggests that the molecules in a substance are always in continuous random motion.
- ◆ When molecules close to each other, the attractive forces between them become strong.
- ◆ The change in force between molecules may cause change of state.
- ◆ Boyle's law describes the pressure - volume relationship of a gas.
- ◆ The pressure and volume of a gas are inversely proportional to each other.
- ◆ Mathematically " $P_1V_1 = P_2V_2$ ".



CONCEPT MAP



End of Unit Questions

Section (A)

- ### Section (A) Multiple Choice Questions (MCQs)



7. A gas in a container of fixed volume is heated. What happens to the molecules of the gas?
- They collide less frequently.
 - They expand.
 - They move faster.
 - They move further apart.
8. In a liquid, some energetic molecules break free from the surface even when the liquid is too cold for bubbles to form. What is the name of this process?
- boiling
 - condensation
 - convection
 - evaporation
9. What happens to the molecules of a gas when the gas changes into a liquid?
- They move closer and lose energy.
 - They move closer and gain energy.
 - They move apart and lose energy.
 - They move apart and gain energy.
10. A substance has a melting point of -17°C and a boiling point of 117°C . In which state does the substance exist at -10°C and at 110°C ?

	at -10°C	at 110°C
a	Solid	liquid
b	solid	gas
c	liquid	liquid
d	liquid	gas

**Section (B) Structured Questions****Kinetic molecular model of matter**

1.
 - a) "The particles are free to move within the material, has a fixed volume but takes up the shape of its container", which state of matter is being described here?
 - b) Write a similar description of the particles that make up a solid.
 - c) Write down any two properties of a solid.
2.
 - a) Why the kinetic model of matter is called kinetic?
 - b) In which state of matter the molecules are widely separated?
 - c) In which state of matter the molecules are most closely packed?
 - d) In which state of matter molecules can move freely at high speed?
3.
 - a) By using kinetic molecular theory explain why we can walk through air, swim through water but can not walk through a solid wall.
 - b) In which state of matter do the molecules have minimum kinetic energy?
 - c) Which state of matter is highly incompressible?

Forces and Kinetic Theory

4. A sample of a gas in a sealed test tube is cooled. Describe what happens to:
 - a) The size of the molecules.
 - b) The speed at which molecules move.
 - c) The number of the molecules.
 - d) The pressure inside the tube.
 - e) The state of the gas.



5. An inflated car tyre is considered to have a constant volume, regardless of any changes in temperature or pressure. Use the kinetic molecular theory of gases to answer the following:
- How does air in the tyre exert a pressure on the walls of the tyre?
 - Why is the pressure the same at all points on the inside wall of the tyre?
 - More air is pumped into the tyre while the temperature is kept constant until there are twice as many molecules as before. Explain why you would expect the pressure to be doubled.
6. Describe the following:
- What happens to the motion of the molecules of a gas when it cools down?
 - What happens to the motion of a liquid when it cools down?

Gases and Kinetic theory

7. The pressure on 9cm^3 of oxygen gas is doubled at a fixed temperature. What will its volume become?
8. A container holds 30m^3 of air at a pressure of 150000Pa . If the volume changed to 10m^3 by decreasing load on the piston. What will the pressure of the gas become? Assume that its temperature remains constant.
9. Air at atmospheric pressure of 760 mm of Hg is trapped inside a container available with a moveable piston. When the piston is pulled out slowly so that the volume is increased from 100dm^3 to 150dm^3 , the temperature remaining constant. What will be the pressure of the air becomes?

Unit - 8

ENERGY SOURCES AND TRANSFER OF ENERGY

Energy is the most important physical concept which is studied in all sciences. work is closely related with energy which provides a link between force and energy. Work, energy and power have special meaning in physics.

Students Learning Outcomes (SLOs)

After learning this unit students should be able to:

- Define work and its SI unit.
- Calculate work done using equation (Work = force $\times$ distance).
- Define kinetic energy and potential energy.
- Use Kinetic Energy ($E_k = \frac{1}{2} mv^2$) and potential energy ($E_p = mgh$); to solve problems.
- Describe the processes by which energy is converted from one form to another with reference to fossil fuel energy, hydroelectric generation, solar energy, nuclear energy, geothermal energy, wind energy, biomass energy and tidal energy.
- Differentiate energy sources as non renewable and renewable energy sources with examples of each.
- Define efficiency of a working system and calculate the efficiency of engine.
- conversion using the formula efficiency = energy converted into the required form / total energy input.
- Explain why a system cannot have an efficiency of 100%.
- Define power and calculate power from the formula Power = work done / time taken.
- Define the unit of power "watt" in SI and its conversion with horse power.



What source of energy is more beneficent? Why we face the shortage of petroleum and gas in our country? Why the people are replacing electric energy by solar energy? Why government is focusing its attention on the use of tidal, solar and wind energy? Why the waste material should be buried in Earth? After learning this unit you will be able to answer these questions and some other similar questions.

8.1 WORK

Generally, work refers to perform some task or job. But in physics work has different meaning. For example: A tailor stitching a suit, a shopkeeper selling fruits at his shop, a women cooking in her kitchen are all considered as "doing work" in daily life but in physics work has a proper meaning i.e. "work is done only when a force makes something to move; Fig (8.1)(a,b)". Thus work can be define as

The amount of work is the product of force and the distance moved in the direction of force or against the direction of force.

Units of Work

The S.I unit of work is Joule other units of work can be Foot, Pound, Erg.

$$1 \text{ Joule} = 1 \text{ Nm}$$

Suppose a constant force "F" acts on a body and motion takes place in a straight line in the direction of force then work done is equal to the product of magnitude of force "F" and the distance "d" through which the body moves.

$$W = Fd \cos \theta \dots \dots \dots (8.2)$$

The force "F" however may not act in the direction of motion of the body instead it makes some angle "θ" with it; Fig (8.2). In that case, we define the

Do You Know! Force is an agent which tends to change the state of an object.

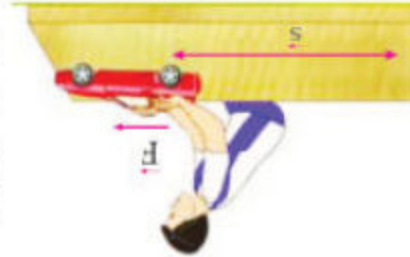


Fig (8.1) (a)
Demonstration of work done

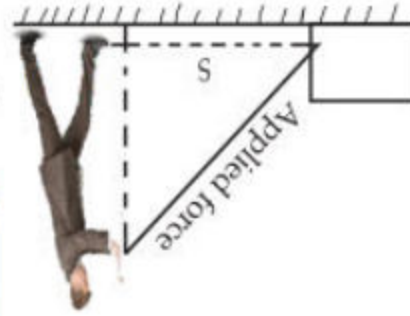


Fig (8.1) (b)
Demonstration of work done



work by the force as the product of the component of the force along the line of motion and the distance "d"; the body moves along that line, i.e.

Suppose a constant force "F" acts on a body

Work = (component of force) · (distance)

$$W = (F \cos \theta) d$$

$$W = (F \cos \theta) d$$

$$\theta = 0 \Rightarrow \cos \theta = 1$$

$$\text{then Work} = W = Fd$$

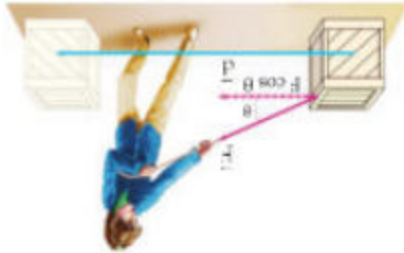


Fig (8.2) (a)
Force making some
angle (θ) with the
direction of motion

Find the work done when a force of 50N is applied to move a trolley at a shopping mall through a distance of 200m. Assume the angle to be of 0° between the force and the distance the trolley moved.

Solution

Step 1: Write down known quantities and quantities to be found.

$$F = 50\text{N}$$

$$d = 200\text{m}$$

$$\theta = 0^\circ$$

$$W = ??$$

Step 2: Write down formula and rearrange if necessary

$$W = F \cdot d$$

$$W = Fd \cos \theta$$

Step 3: Put the values in formula and calculate:

$$W = 50\text{N} \times 200\text{m} \times \cos 0^\circ$$

$$W = 10000\text{J}$$

Hence, the work done is 10000 Joules or 10 KJ.

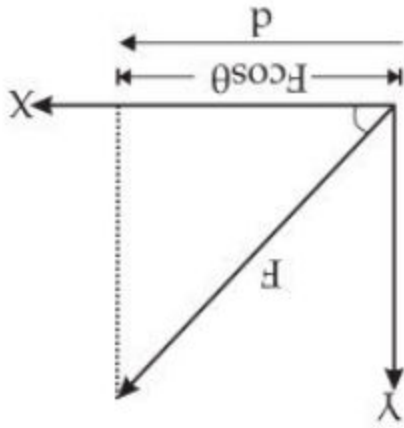


Fig (8.2) (b)
Force making some
angle (θ) with the
direction of motion

Self Assessment Questions:

- Q1: Write down the names of any three units of work.
Q2: According to the definition of work in physics, Urwa did not perform any work if she made and assignment on her laptop in three hours. Why?
Q3: At what angle between force and displacement the work done by a body will be maximum?

Energy

Energy is define as

The ability to do work.

the S.I unit of energy is joule (J). There are many forms of energy. Some of them are discussed below:

8.2 Energy Forms

Kinetic Energy

Kinetic energy of a body defined as:

Energy possessed by an object due to its motion is called kinetic energy.

The S.I unit of kinetic energy is joule.

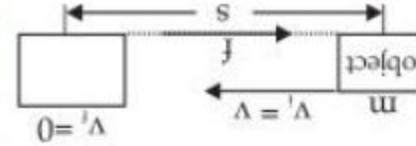
It is also defined as "The work required to accelerate a body of a given mass from rest to its stated velocity". A moving body maintains its kinetic energy unless its speed changes.

Mathematically kinetic energy is given as:

$$K.E = \frac{1}{2}mv^2 \dots\dots\dots(8.3)$$

As we know that kinetic energy is due to the motion of object. Therefore for an object of mass m moving with speed (v) kinetic energy depends upon:

- ◆ the mass m of the object- the greater the mass, the greater its (K.E).
- ◆ the speed (v) of the object- the greater the speed, the greater the (K.E).



Do You Know! Sailing boat, moving car, running and walking are example of kinetic Energy.





Worked Example 2

A ball of mass 400 gram, strikes the wall of velocity 4m/sec. How much is the kinetic energy of the ball at the time it strikes the wall?

Solution

Step 1: Write down known quantities and quantities to be found.

$$m = 400 \text{ gram} = \frac{400}{1000} \text{ kg} = 0.4 \text{ kg}$$

$$v = 4 \text{ ms}^{-1}$$

$$K.E = ?$$

Step 2: Write down formula and rearrange if necessary

$$K.E = \frac{1}{2} mv^2$$

Step 3: Put the values in formula and calculate

$$K.E = \frac{1}{2} \times 0.4 \text{ kg} \times (4 \text{ ms}^{-1})^2$$

$$K.E = 3.2 \text{ J}$$

Hence, Kinetic Energy is possessed by the ball 3.2 joules.

Potential Energy

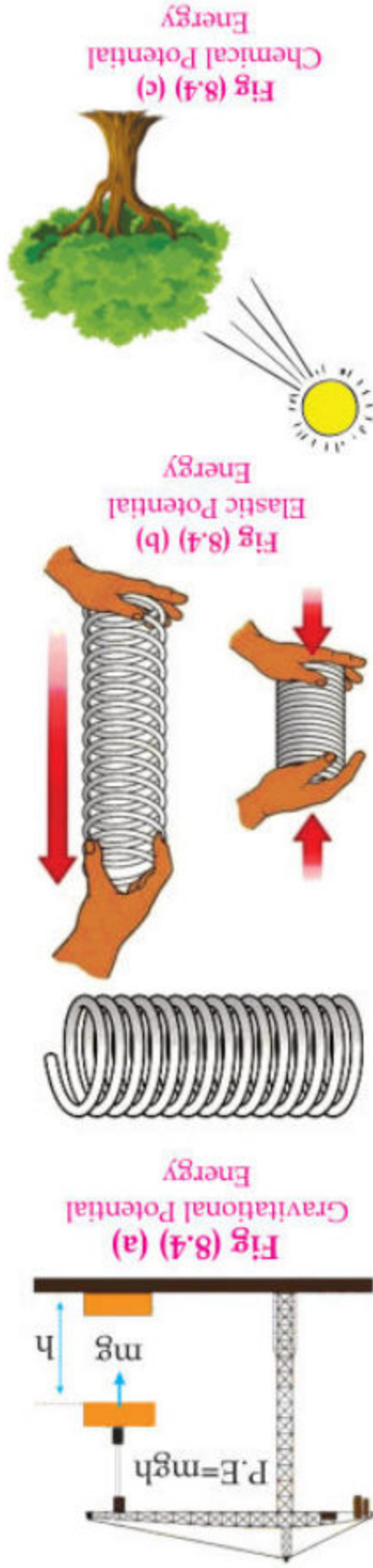
Potential energy of a body is defined as:

The energy that a body possesses by virtue of its position, shape or state of a system.

There are different types of potential energy. Like gravitational potential energy, elastic potential energy and chemical potential energy; Fig (8.4) (a, b, c).

For Example:

- ◆ A body raised to a height "h" above the ground has gravitational potential energy.
- ◆ A stretched spring has elastic potential energy due to its stretched position (condition).





◆ The energy stored in the plants that we eat is chemical potential energy.

S.I. unit of potential energy is Joule (J).

It is also defined as the work done stored in a body in lifting it to a height "h". The potential energy changes only when its position relative to ground changes; otherwise it remains same.

Mathematically potential energy is given as

$$P.E = mgh \quad \dots\dots\dots (8.4)$$

Derivation of Gravitational Potential Energy (P.E=mgh)

To derive the expression for gravitational potential energy, let us consider an object of mass "m" which is raised up through height "h" from the ground; Fig (8.5). The work done in lifting it to height "h" is stored in it as its gravitational potential energy "P.E", i.e.

P.E = Work done

P.E = W

P.E = F.d

P.E = (mg) . h

Therefore equation becomes:

$$P.E = mgh \quad \dots\dots\dots (8.4)$$

Worked Example 3

A ball of mass 50 gram is raised to a height of 7m from the ground. Calculate its gravitational potential energy.

Solution

Step 1: Write down known quantities and quantities to be found.

$$m = 50\text{gm} = \frac{50}{1000}\text{kg} = 0.05\text{kg}$$

Do You Know!
A book lying on the table and the water stored in a dam have potential energies.

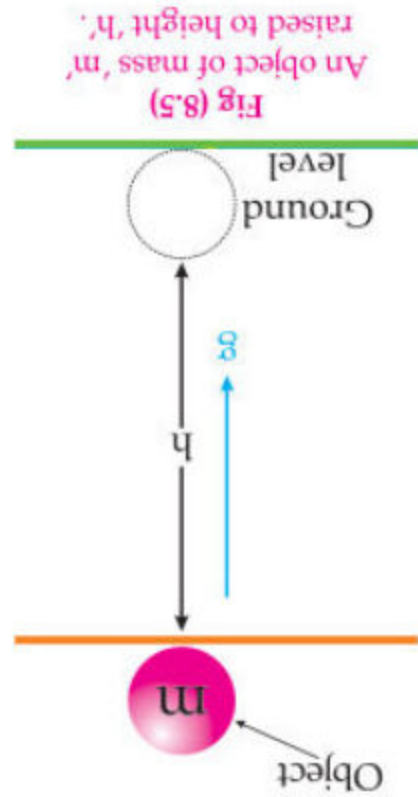


Fig (8.5)
An object of mass 'm' raised to height 'h'.



Fig (8.6)
Fossil fuel energy

Self Assessment Questions:

Q4: A car of mass 50kg moving with velocity 10ms^{-1} in the direction of force. Calculate its kinetic energy.

Q5: A body of mass 10kg is dropped from a height of 20m on the ground. What will be its potential energy, if $g=9.8\text{m/sec}^2$?

Q6: Give the energy changes when a ball is dropped from a height of 7m to the ground.

8.3 CONVERSION OF ENERGY

Energy neither be created nor it can be destroyed but it can be converted from one form to other form. This is called law of conservation of energy.

1. Fossil Fuel Energy

Fossil fuel energy is formed from decayed plants and animals that have been converted to crude oil, coal, natural gases or heavy oils by exposure to heat and pressure in the Earth's crust over hundreds of millions of years; Fig (8.6).

Conversion of Energy from one form to another



Fossils fuels have stored chemical energy. This energy is converted by oxidation through burning. Thus on burning a fossil fuel like charcoal, produce heat energy and light energy; Fig (8.7).

ii. Hydroelectric Energy

Hydro electricity is the term referring to electricity generated by hydro power by using gravitational force of falling or flowing water; Fig (8.8).
Most common type of hydro electric power plants uses a dam on a river to store water in a reservoir. Water releases from the reservoir flows through a turbine, spinning it, which in turn runs a generator to produce electricity.



Fig (8.7)
Burning charcoal



Fig (8.8)
Hydroelectric energy

The energy radiated from the sun is known as solar energy. This is the most available source of energy throughout Pakistan. There are many devices which are capable of absorbing solar energy, which is then converted into electrical energy or heat energy. These devices may be photovoltaic solar panels and solar cells. Which convert the sun rays into electricity for different uses; Fig (8.9)(a). Also solar heaters are used to convert solar energy "sun rays" into heat energy to heat water tanks and indoor spaces; Fig (8.9)(b).

iv. Nuclear Energy

The energy released during a nuclear reaction such as fission or fusion reaction. All radioactive materials store nuclear energy. For example Uranium,



Fig (8.9) (a)
Solar energy (Solar Panel)



Fig (8.9) (b)
Solar energy (Solar heater)



Radium etc. It is released from the nucleus in the form of radiation in addition to heat and light. A nuclear power plant utilize nuclear energy to produce steam to turn a turbine and generate electricity; Fig (8.10).



Fig (8.10)
Nuclear energy

Geothermal energy is stored in the Earth as its natural heat. Deep in the Earth, there is hot molten part called magma. Water close to magma changes to steam due to high temperature. This thermal energy is conducted to the surface of Earth. This energy is called geothermal energy; Fig (8.11)(a).



Fig (8.11)(a)
Geothermal energy

A geothermal power plant utilizes geothermal energy to drive an electrical generator; Fig 8.(11) (b). Geothermal well can be built by drilling deep near hot rocks at different places, where hot molten or magma is very close, water is then pushed down into the well. The rocks quickly heat the water and change it into steam. The steam is used for heating purpose or to generate electricity.



Fig (8.11)(b)
Geothermal power plant

vi. Wind Energy

The energy obtained by the wind is called wind energy. It is generated by wind mills (Fig 8.12). A wind mill consists of a turbine which rotates due to wind. Kinetic energy is produced due to the motion of turbine. Wind turbines convert this kinetic energy into the mechanical power. A generator converts that mechanical power into electricity.



Fig(8.12)
Wind energy

- ◆ It is being used as source of energy for sailing ships in oceans.
- ◆ It is being used by wind mills to pump water.